

Lab 3: Self-Attention by Hand

One head by hand, a heatmap, then a causal mask — and why zero is the wrong fill.

Work in a **Jupyter** notebook with **PyTorch**. Do notes **3.1–3.4** first.

First time you implement this. The lecture notes this week already answered what the method is, why it exists, the architecture, the formula, and the tradeoffs. Read that first-time block before these exercises. This lab is the first time you **compute** it, not the first definition.

```
import torch
import torch.nn.functional as F
import matplotlib.pyplot as plt
```

The algebra is note 3.2. The causal mask is the GPT fork in note 3.4.

1. One head, by hand

Four tokens, $d = 2$. You will compute attention without `nn.MultiheadAttention`.

```
torch.manual_seed(0)
X = torch.tensor([
    [1.0, 0.0], # The
    [0.0, 1.0], # cat
    [1.0, 1.0], # sat
    [0.5, 0.2], # on
])
Wq = torch.eye(2)
Wk = torch.eye(2)
Wv = torch.eye(2)
```

1. Compute Q, K, V . With these weights they equal X .
 2. Scores $S = QK^T / \sqrt{2}$. Print S . Why $\sqrt{2}$, not 2?
 3. $A = \text{softmax}(S, \text{dim} = -1)$. Rows must sum to 1. Heatmap A with token names The, cat, sat, on.
 4. Output $Y = AV$. Which token's output is closest (cosine) to cat? Guess before you compute. Confirm you mixed **values**, not keys.
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2. A causal mask

1. Build a mask M with 0 on and below the diagonal and $-\infty$ strictly above. Add it to S before softmax. Print the new A . The last token may see everyone; the first token may see only itself.
 2. Briefly: replace $-\infty$ by 0 and softmax again. Why is that the wrong mask?
 3. In one sentence: why would a GPT-style model be cheating without this mask?
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3. Library check

```
attn = torch.nn.MultiheadAttention(embed_dim=2, num_heads=1, batch_first=True)
```

You do not need to train it. Copy `attn.in_proj_weight` is optional. The point of (1)–(2) is the algebra. Write four sentences: query vs key vs value, what a row of A is, why Week 2's hidden state is no longer the only memory, and whether this four-token map is GPT-style or BERT-style before you add the mask.